12.7 Who Meets Whom in a Knockout

WORKSHEET

The question

STEP II Specimen paper, Question 16

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s2-q16

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself How many matches are played in a knockout tournament with 2^n players? (Each match eliminates one player.)

STAGE 2 The key symmetry

Ask yourself The draw in each round is random and every match is a fair coin toss. Does the process treat any pair of players differently from any other?

STAGE 4 At any stage

Ask yourself X and Y can meet at most once. How does that let you add probabilities over matches?

Try this next

1. **Round by round.** Show directly that the probability that X and Y meet in round k is $\frac{2^{n-k}}{\binom{2^n}{2}}$, and check that these add up to $\frac{1}{2^{n-1}}$.
2. **A fixed bracket.** If instead the bracket is fixed in advance (as in the figure) but all players are placed in it at random, does the answer change?
3. **A strong player.** Suppose player X wins every match. What is now the probability that X meets Y?
4. **A league.** In a league of $2n$ teams where every pair meets once, how many matches are there? What is the analogous "probability of meeting"?

Further reading

- **Frederick Mosteller, *Fifty Challenging Problems in Probability*** (Dover), which includes tournament problems.
- **David Spiegelhalter, *The Art of Statistics*** (Pelican), on modelling and forecasting.
- **Joseph Blitzstein & Jessica Hwang, *Introduction to Probability*** (CRC Press; free online), on indicators and symmetry.

The full guided solution, and what the question leads to at university, is in Section 12.7:
https://beyondstep.keithmcnulty.org/ch12/knockout.html